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(54) THIAZOLIDONE SPIRO PYRIMIDINE TRIONE COMPOUND, PREPARATION METHOD THEREFOR AND USES THEREOF

THIAZOLIDON-SPIRO-PYRIMIDIN-TRION-VERBINDUNG, VERFAHREN ZU IHRER HERSTELLUNG UND VERWENDUNGEN DAVON

COMPOSÉ DE THIAZOLIDONE-SPIROPYRIMIDINETRIONE, SON PROCÉDÉ DE PRÉPARATION ET SES UTILISATIONS

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- **BASARAB, G.S. et al.: "Discovery of Novel DNA Gyrase Inhibiting Spiropyrimidinetriones: Benzisoxazole Fusion with N-Linked Oxazolidinone Substituents Leading to a Clinical Candidate (ETX0914)", Journal of Medicinal Chemistry, vol. 58, no. 15, 9 July 2015 (2015-07-09), pages 6264-6282, XP055555089, ISSN: 0022-2623**

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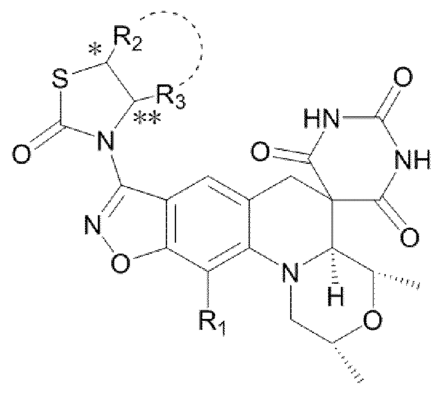


(I)

wherein, R_1 is hydrogen, halogen or cyano;

R_2 and R_3 are each independently hydrogen, phenyl, substituted or unsubstituted C_1-C_6 alkyl, or $=CH_2$, and the term "substituted" refers to a substitution with a substituent selected from the group consisting of C_1-C_6 alkoxy, phenyl, halogen, $-N_3$, $-S(O_2)C_1$-C_6 alkyl, $-NHCOC_1$-C_6 alkyl, $-CONHC_1$-C_6 alkyl, $-OR_4$, C_1-C_6 alkylthio, C_2-C_6 alkenyl, C_2-C_6 alkynyl; or R_2, R_3 and the attached carbon atoms together form a 4-7 membered aliphatic ring;
R_4 is hydrogen, or C_1-C_6 haloalkyl;
* and ** each independently represent a racemic, S-type or R-type.

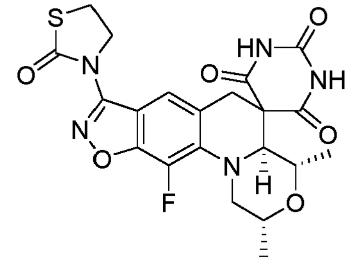
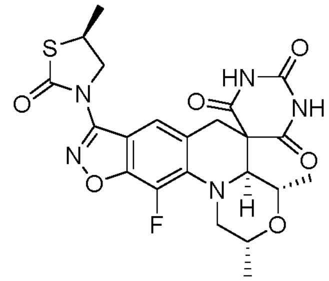
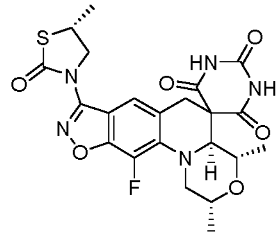
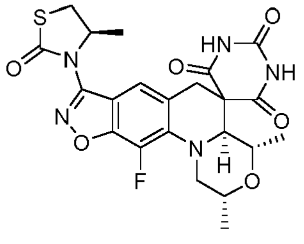
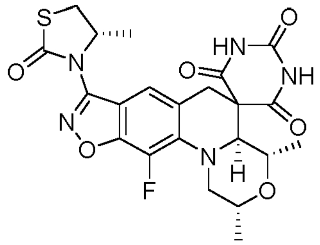
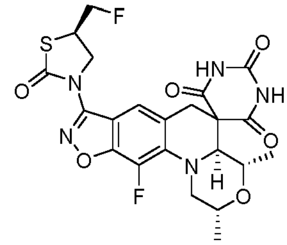
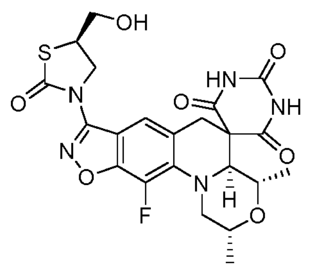
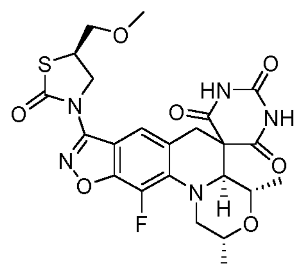
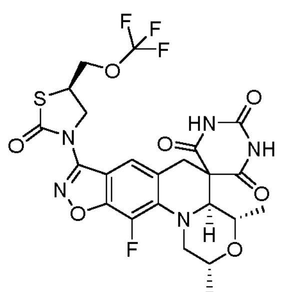
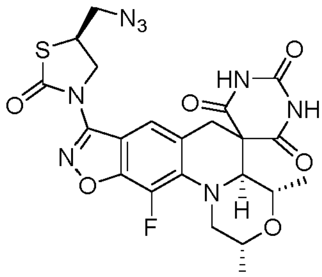
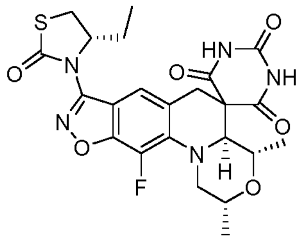
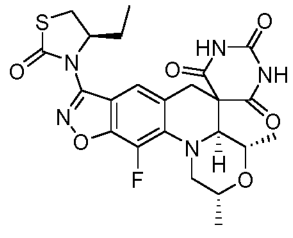
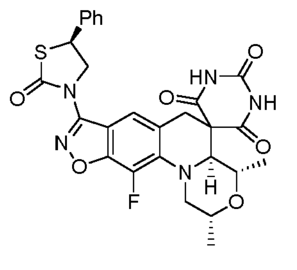
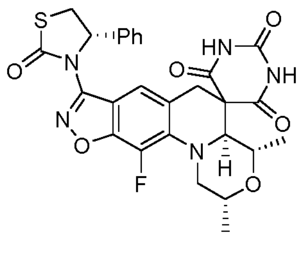
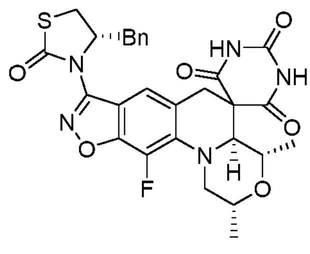
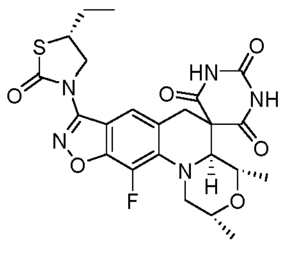
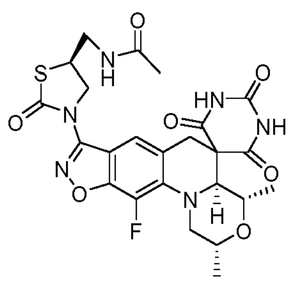
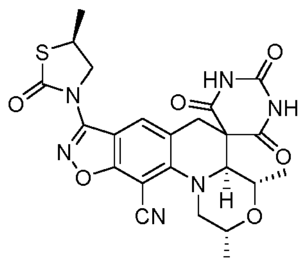
[0013] The general formula (I) of the present invention indicates that the compound contains at least three chiral centers, and there are enantiomers and diastereomers. For enantiomers, two enantiomers can be obtained by general chiral resolution or asymmetric synthesis. Diastereomers can be separated by fractional recrystallization or chromatographic separation. The compound represented by general formula (I) of the present invention includes any of the above isomers or a mixture thereof.
[0014] In another preferred embodiment, R_1 is fluorine, chlorine or cyano.
[0015] In another preferred embodiment, R_2 is hydrogen, phenyl, substituted or unsubstituted C_1-C_4 alkyl, or $=CH_2$, wherein the term "substituted" refers to a substitution with a substituent selected from the group consisting of halogen, $-OR_4$, C_1-C_4 alkoxy, $-N_3$, $-NHCOC_1$-C_4 alkyl, $-S(O_2)C_1$-C_4 alkyl; R_4 is hydrogen or C_1-C_4 haloalkyl;

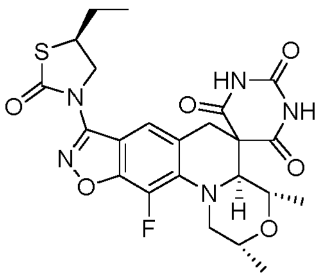
R_3 is hydrogen, substituted or unsubstituted C_1-C_4 alkyl, or phenyl, wherein the term "substituted" refers to a substitution with a substituent selected from the group consisting of: phenyl;
or R_2 and R_3 together with the attached carbon atoms form a 5-6 membered aliphatic ring.

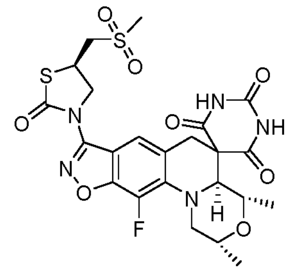
[0016] In another preferred embodiment, R_2 is hydrogen, phenyl, substituted or unsubstituted C_1-C_3 alkyl, or $=CH_2$, and the term "substituted" refers to a substitution with a substituent selected from the group consisting of fluorine, chlorine, $-OR_4$, C_1-C_3 alkoxy, $-N_3$, $-NHCOC_1$-C_3 alkyl, $-S(O_2)C_1$-C_3 alkyl; R_4 is hydrogen or trifluoromethyl;

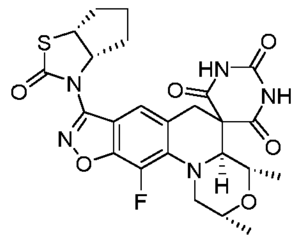
R_3 is hydrogen, C_1-C_3 alkyl, phenyl, or benzyl;
or R_2 and R_3 together with the attached carbon atoms form a 5-6 membered aliphatic ring.

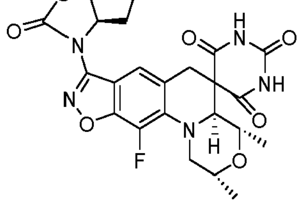
[0017] In another preferred embodiment, at most one of R_2 and R_3 is hydrogen; or R_2 and R_3 together with the attached carbon atoms form a 5-6 membered aliphatic ring.
[0018] In another preferred embodiment, at least one of R_2 and R_3 is hydrogen; or R_2 and R_3 together with the attached carbon atoms form a 5-6 membered aliphatic ring.
[0019] In another preferred embodiment, R_4 is hydrogen or trifluoromethyl.
[0020] In another preferred embodiment, the compound is:

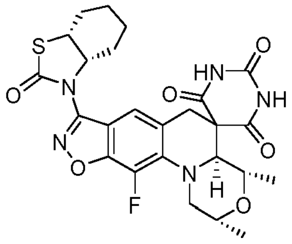



















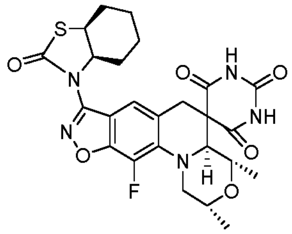












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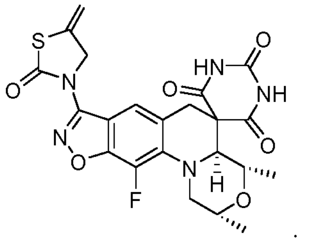
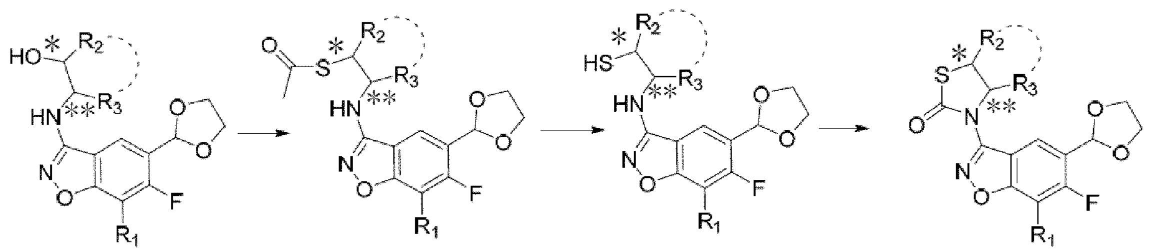


[0021] In the second aspect of the present invention, it provides a pharmaceutical composition comprising the compound or enantiomer, diastereomer, racemate and mixture thereof, or pharmaceutically acceptable salt thereof in the first aspect; and

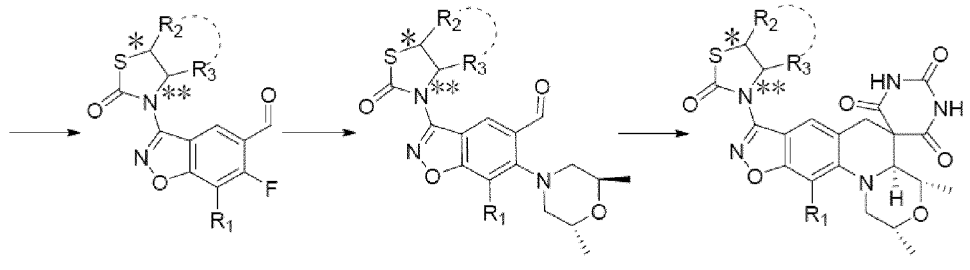
a pharmaceutically acceptable carrier or excipient.

[0022] The invention provides a novel benzoisoxazole spiro pyrimidine trione compound, which can be used alone, or is mixed with pharmaceutically acceptable excipients (such as excipients, diluents, etc.) and formulated into tablets, capsules, granules or syrups for oral administration. The pharmaceutical composition can be prepared according to a conventional method in pharmaceuticals.

[0023] In the third aspect of the present invention, it provides a method for preparing the compound in the first aspect, comprising the following steps:



Ia Ib Ic Id



Ie If I

(i) subjecting an intermediate Ia and thioacetic acid to Mitsunobu reaction, thereby forming an intermediate Ib;
(ii) subjecting the intermediate Ib and sodium hydroxide to hydrolysis reaction, thereby forming an intermediate Ic;
(iii) subjecting the intermediate Ic and N,N'-carbonyldiimidazole to nucleophilic substitution reaction, thereby forming an intermediate Id;
(iv) subjecting the intermediate Id and hydrochloric acid to a deprotection reaction, thereby forming an intermediate Ie;
(v) subjecting the intermediate Ie and 2R,6R-dimethylmorpholine to nucleophilic substitution reaction, thereby forming an intermediate If;
(vi) reacting the intermediate If with barbituric acid, thereby forming a compound represented by formula I,

in each formula, *, **, R_1, R_2, and R_3 are defined as above.

[0024] In another preferred embodiment, the intermediate Ia is reacted with thioacetic acid via Mitsunobu reaction to form the intermediate Ib.

[0025] In another preferred embodiment, the intermediate Ib and sodium hydroxide are subjected to a hydrolysis reaction to form the intermediate Ic.

[0026] In another preferred embodiment, the intermediate Ic and N, N'-carbonyldiimidazole undergo a nucleophilic substitution reaction to form the intermediate Id.

[0027] In another preferred embodiment, the intermediate Id is reacted with hydrochloric acid to remove the protective group to form the intermediate Ie.

[0028] In another preferred embodiment, the intermediate Ie and 2R,6R-dimethylmorpholine undergo a nucleophilic substitution reaction to form an intermediate If.

[0029] In the fourth aspect of the present invention, it provides a compound or enantiomer, diastereomer, racemate and mixture thereof, or a pharmaceutically acceptable salt thereof in the first aspect or a pharmaceutical composition in the second aspect for use in treating a bacterial infectious disease.

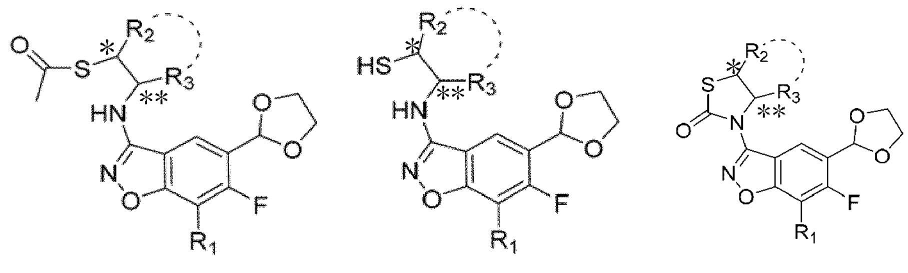
[0030] In another preferred embodiment, the infectious disease is an infectious disease caused by a multidrug-resistant bacterium.

[0031] In another preferred embodiment, the infectious disease is an infectious disease caused by Gram-positive bacterium.

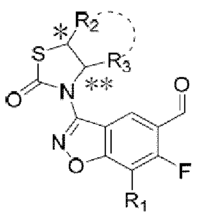
[0032] In another preferred embodiment, the multidrug-resistant bacterium is selected from the group consisting of MRSA, MSSA, MRSE, MSSE, PRSP and Spy.

[0033] A method for inhibiting bacteria *in vitro,* comprising administering to a subject or the environment the compound or enantiomer, diastereomer, racemate and mixture thereof, or a pharmaceutically acceptable salt thereof in the first aspect or the pharmaceutical composition in the second aspect is also disclosed.

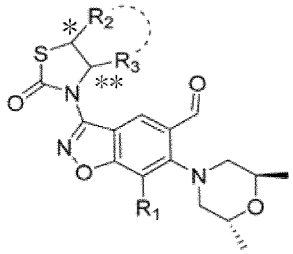
[0034] In the fifth aspect of the present invention, it provides an intermediate of a compound of formula I, the structure of which is represented by Formula Ib, Ic, Id, Ie or If:



Ib Ic Id



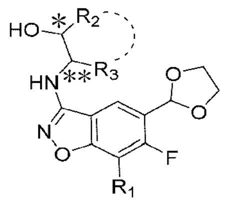
Ie



If

in each formula, R_1, R_2, and R_3 are defined as above.

[0035] An intermediate of a compound of formula I which is represented by Formula Ia



Ia

is also disclosed herein.

[0036] The compound of the present invention has much better antibacterial activity *in vitro* and *in vivo* and drug metabolism properties than AZD0914, and is very suitable as a novel antibacterial medicine for the antibacterial treatment of human or animal infections.

Description of embodiments

[0037] After extensive and intensive research, the inventors of the present application have firstly developed a class of benzoisoxazole spiro pyrimidine trione compounds having novel structure. It is found that they have much better antibacterial activity *in vitro* and *in vivo* and drug metabolism properties than AZD0914, and are very suitable as a novel antibacterial drug for the antibacterial treatment of human or animal infections. Based on this, the present invention has been completed.

Terms

[0038] In the present invention, the term "C_1-C_6 alkyl" refers to a linear or branched alkyl having 1 to 6 carbon atoms, and includes, without limitation, methyl, ethyl, propyl, isopropyl, butyl and the like; and the term "C_1-C_4 alkyl" has a similar meaning. The term "C_2-C_6 alkenyl" refers to a linear or branched alkenyl with one double bond having 2 to 6 carbon atoms, and includes but is not limited to vinyl, propenyl, butenyl, isobutenyl, pentenyl and hexenyl. The term "C_2-C_6 alkynyl" refers to a straight or branched alkynyl having 2 to 6 carbon atoms and containing a triple bond, including, without limitation, ethynyl, propynyl, butynyl, isobutynyl, pentynyl and hexynyl. The term "C_1-C_6 haloalkyl" refers to an alkyl substituted with one or more halogen atoms, such as $-CH_2F$, $-CF_3$, $-CH_2CHF_3$, and the like.

[0039] In the present invention, the term "C_1-C_6 alkoxy" refers to a straight or branched alkoxy having 1 to 6 carbon atoms, and includes, without limitation, methoxy, ethoxy, propoxy, isopropoxy, butoxy and the like; and the term "C_1-C_4 alkoxy" has a similar meaning.

[0040] In the present invention, the term "4- to 7-membered aliphatic ring" refers to a cycloalkyl having 4 to 7 carbon atoms on the ring, and includes, without limitation, cyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl and the like; and the term "5- to 6-membered aliphatic ring" has a similar meaning.

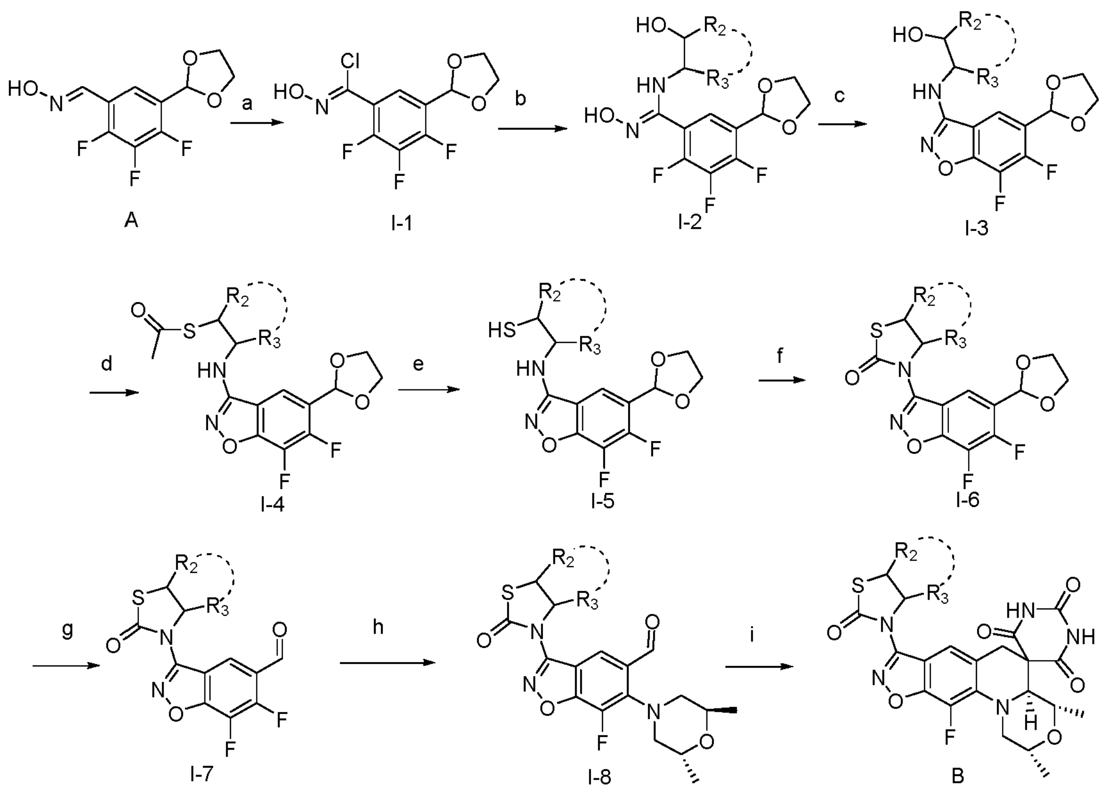
[0041] The "pharmaceutically acceptable salt" includes a pharmaceutically acceptable base addition salt, which includes, but is not limited to, a salt of inorganic base such as sodium salt, potassium salt, calcium salt, magnesium salt and the like, and includes, but is not limited to, a salt of organic base such as ammonium salt, triethylamine salt, lysine salt, arginine salt, and the like. These salts can be prepared by methods known in the art.

Preparation method

[0042] The compound represented by the general formula (I) of the present invention can be prepared by the following method, but the conditions of the method, such as the reactant, solvent, base, amount of the compound used, reaction temperature, reaction time, etc. are not limited to the following explanations. The compounds of the present invention can also be conveniently prepared by optionally combining various synthetic methods described in this specification or known in the art, and such combinations can be easily performed by those skilled in the art to which the present invention belongs.

Route 1

[0043] In a preferred embodiment, compounds **1-5**, compounds **11-16**, compound **19**, and compounds **21-24** are prepared according to route 1.



R_2 and R_3 are defined as above.

a. An intermediate **A** [J. Med. Chem, 2015, 58 (15): 6264-6282.] is reacted with N-chlorosuccinimide in a polar aprotic solvent at 10-60 °C for 20-60 minutes to obtain an intermediate **I-1.** The polar aprotic solvent may be 1,4-dioxane, toluene, tetrahydrofuran, or dimethylformamide (DMF). The optimal reaction temperature is room temperature and the optimal reaction time is 30 min.

b. The intermediate **I-1** is reacted with different primary amines respectively in polar aprotic solvents at 0 °C for 30 min to form a corresponding intermediate **I-2** via nucleophilic substitution. The primary amines are various primary amines that meet the requirements and the polar aprotic solvent may be 1,4-dioxane, toluene, tetrahydrofuran, or dimethylformamide (DMF).

c. The intermediate **I-2** is reacted in a polar aprotic solvent at 60 °C for 4h (hours) in the presence of cesium carbonate to generate a corresponding intermediate **I-3.** The polar aprotic solvent may be 1,4-dioxane, toluene, tetrahydrofuran, or dimethylformamide (DMF).

d. The intermediate **I-3** is reacted with thioacetic acid in the presence of an active intermediate formed by diisopropyl azodicarboxylate and triphenylphosphine in a polar aprotic solvent at room temperature for 2 h, thereby forming a corresponding intermediate **I-4**. The polar aprotic solvent may be 1,4-dioxane, toluene, tetrahydrofuran, or dimethylformamide (DMF).

e. The intermediate **I-4** is reacted in the presence of sodium hydroxide and dithiothreitol in a solvent at 0 °C for 1 h to form a corresponding intermediate **I-5**. The solvent may be methanol, ethanol, 1,4-dioxane, tetrahydrofuran, or dimethylformamide (DMF).

f. The intermediate **I-5** is reacted with N,N'-carbonyldiimidazole under the catalysis of 4-dimethylaminopyridine in a polar aprotic solvent at 80 °C for 4h to generate a corresponding intermediate **I-6**. The polar aprotic solvent may be 1,4-dioxane, toluene, tetrahydrofuran, or dimethylformamide (DMF).

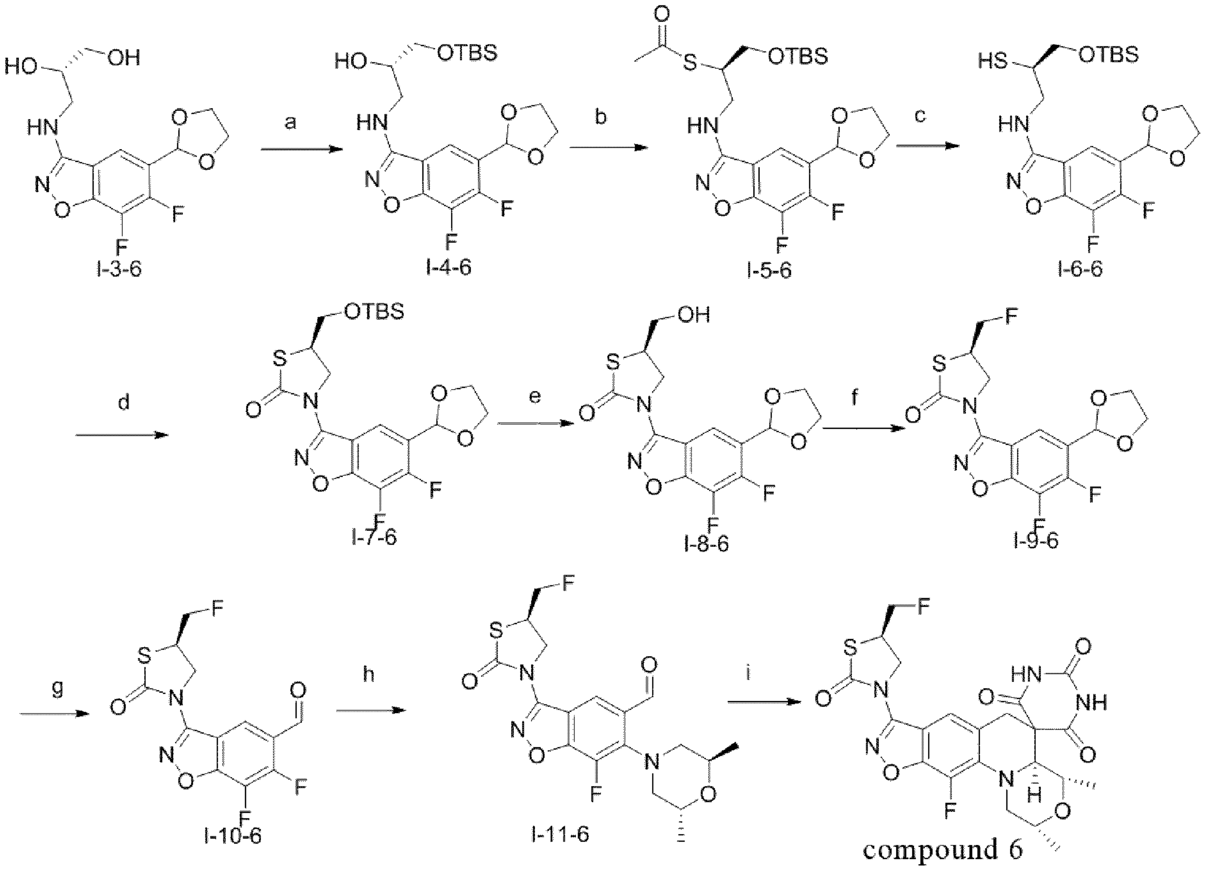
g. The intermediate **I-6** and hydrochloric acid are reacted in a solvent at room temperature for 2h to form a corresponding intermediate **I-7.** The solvent may be 1,4-dioxane, tetrahydrofuran, or dimethylformamide (DMF).

h. The intermediate **I-7** is reacted with 2R,6R-dimethylmorpholine in the presence of an organic base in a polar aprotic solvent at 90 °C for 15 hours to form a corresponding intermediate **I-8.** The organic base may be triethylamine or N,N-diisopropylethylamine, and the polar aprotic solvent may be acetonitrile, 1,4-dioxane, tetrahydrofuran, or dimethylformamide (DMF).

i. The intermediate **I-8** is reacted with barbituric acid in a mixed solvent of acetic acid and water at 110 °C for 5 hours to form the corresponding compound **B**.

Route 2

[0044] In a preferred embodiment, compound 6 is prepared according to route 2.



a. An intermediate **I-3-6** [prepared by route 1] is reacted with tert-butyldimethylsilyl chloride in the presence of an organic base in a polar aprotic solvent at room temperature for 2 hours to produce an intermediate **I-4-6.** The organic base may be imidazole, and the polar aprotic solvent may be 1,4-dioxane, tetrahydrofuran, or dimethylformamide (DMF).

b. The intermediate **I-4-6** is reacted with thioacetic acid in the presence of an active intermediate formed by diisopropyl azodicarboxylate and triphenylphosphine in a polar aprotic solvent at room temperature for 2 h to form a corresponding intermediate **I-5-6.** The polar aprotic solvent may be 1,4-dioxane, toluene, tetrahydrofuran, or dimethylformamide (DMF).

c. The intermediate **I-5-6** is reacted in the presence of sodium hydroxide and dithiothreitol in a solvent at 0 °C for 1 h to form an corresponding intermediate **I-6-6.** The solvent can be methanol, ethanol, 1,4-dioxane, tetrahydrofuran, or dimethylformamide (DMF).

according to body weight. Each group contained eight mice, half male and half female, and each mouse was injected intraperitoneally with the test bacteria solution with a dose of 0.5 ml per 20 g mouse body weight. 0.5h and 4h after infection, the mice were administered via intragastric administration according to the designed dose, 0.5 ml per 20 g mouse body weight. They were observed for 7-14 days and the death numbers of mice were recorded. Based on the death numbers of mice, half effective dose ED50 and 95% confidence limit were calculated by Bliss method using DAS1.0 software edited by Sun Ruiyuan, et al..

[0374] Since the mice metabolic clearance rate of the control drug AZD0914 was too high and the metabolic properties were not ideal, the mice in AZD0914 control group must be orally administered with a cytochrome P450 inhibitor, 1-aminobenzotriazole (ABT) at a dose of 50 mg/kg two hours before infection. After 12 hours, the mice were orally administered with 50 mg/kg ABT again. The mice in the compound group did not require oral administration of ABT.

[0375] The mice in the infection control group were only infected with bacteria and were not administered with the medicine, and the same volume of saline was administered via intragastric administration after infection. The mice in the blank control group were not infected with bacteria, the same volume of saline was administered via intraperitoneal injection, and the same volume of saline was administered via intragastric administration.

4. Test results

[0376] The *in vivo* antibacterial activity test results of the systemic infection model are shown in Table 7.

Table 7 *In vivo* protective effect of the compounds of the present invention on *S. aureus* MRSA15-3 infected mice (ED_{50})

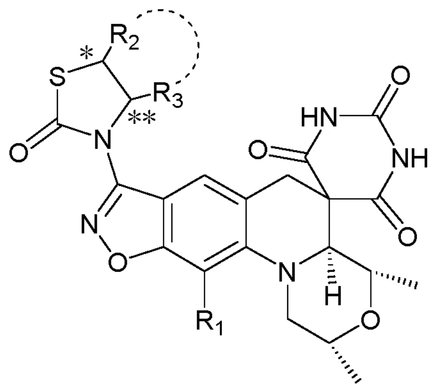
Drug	Administration route	ED_{50} (95% confidence limit) (mg/kg)
Compound 2	i.g. (intragastric administration)	2.54 0.11--3.71
AZD0914	i.g. (intragastric administration)	11.51 7.28--25.79

[0377] It can be seen from Table 7 that oral administration of Compound **2** of the present invention has a four-fold stronger *in vivo* protective effect on *S. aureus* MRSA15-3 infected mice than that of the positive control drug AZD0914, so that the effect is far superior to that of AZD0914.

[0378] In summary, the compounds of the present invention have better drugability than the existing positive control drug AZD0914, and are expected to become a better antibacterial drug.

Claims

1. A compound represented by the general formula I, or an enantiomer, a diastereomer, a racemate and a mixture thereof, or a pharmaceutically acceptable salt thereof,



(I)

wherein, R_1 is hydrogen, halogen or cyano;

R_2 and R_3 are each independently hydrogen, phenyl, substituted or unsubstituted C_1-C_6 alkyl, or $=CH_2$, and the term "substituted" refers to a substitution with a substituent selected from the group consisting of C_1-C_6 alkoxy, phenyl, halogen, $-N_3$, $-S(O_2)C_1$-C_6 alkyl, $-NHCOC_1$-C_6 alkyl, $-CONHC_1$-C_6 alkyl, $-OR_4$, C_1-C_6 alkylthio, C_2-C_6 alkenyl, C_2-C_6 alkynyl; or R_2, R_3 and the attached carbon atoms together form a 4-7 membered aliphatic ring;

R_4 is hydrogen, or C_1-C_6 haloalkyl;

* and ** each independently represent a racemic, S-type or R-type.

2. The compound of claim 1, wherein R_1 is fluorine, chlorine or cyano.

3. The compound of claim 1, wherein R_2 is hydrogen, phenyl, substituted or unsubstituted C_1-C_4 alkyl, or $=CH_2$, and the term "substituted" refers to a substitution with a substituent selected from the group consisting of halogen, $-OR_4$, C_1-C_4 alkoxy, $-N_3$, $-NHCOC_1$-C_4 alkyl, $-S(O_2)C_1$-C_4 alkyl; R_4 is hydrogen or C_1-C_4 haloalkyl;

R_3 is hydrogen, substituted or unsubstituted C_1-C_4 alkyl, or phenyl, wherein the term "substituted" refers to a substitution with a substituent selected from the group consisting of: phenyl;

or R_2 and R_3 together with the attached carbon atoms form a 5-6 membered aliphatic ring.

4. The compound of claim 1, wherein R_2 is hydrogen, phenyl, substituted or unsubstituted C_1-C_3 alkyl, or $=CH_2$, and the term "substituted" refers to a substitution with a substituent selected from the group consisting of fluorine, chlorine, $-OR_4$, C_1-C_3 alkoxy, $-N_3$, $-NHCOC_1$-C_3 alkyl, $-S(O_2)C_1$-C_3 alkyl; R_4 is hydrogen or trifluoromethyl;

R_3 is hydrogen, C_1-C_3 alkyl, phenyl, or benzyl;

or R_2 and R_3 together with the attached carbon atoms form a 5-6 membered aliphatic ring.

5. The compound of claim 1, wherein R_4 is hydrogen or trifluoromethyl.

6. The compound of claim 1, wherein the compound is: